

MIS2002S - Digital Business





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Agenda

Session	Topic
1	Intro To Digital Business Assignment Walkthrough
2	Digital Business Infrastructure Artificial Intelligence (AI) Assignment Walkthrough
3	Digital Business Strategy B2B Digital Business
4	Digital Marketing
5	Digital Business Security Ethical & Legal Issues of Digital Business
6	Individual Assignment Recap Assignment Q&A

Digital Business



Topic 1

Intro To Digital Business

Learning Objectives

- 1.1 Understand why it is important to study e-commerce / e- business
- 1.2 Define e-commerce, understand how e-commerce differs from e-business, identify the primary technological building blocks underlying e-commerce, and recognize major current themes in e-commerce.
- 1.3 Identify and describe the unique features of e-commerce technology and discuss their business significance.
- 1.4 Describe the major types of e-commerce.
- 1.5 Understand the evolution of e-commerce from its early years to today.
- 1.6 Describe the major themes underlying the study of e-commerce.
- 1.7 Identify the major academic disciplines contributing to e-commerce.

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ESSENTIAL DIGITAL HEADLINES

OVERVIEW OF THE ADOPTION AND USE OF CONNECTED DEVICES AND SERVICES

TOTAL
POPULATION



we
are
social

**8.20
BILLION**

URBANISATION

58.1%

UNIQUE MOBILE
PHONE SUBSCRIBERS



Meltwater

**5.78
BILLION**

vs. POPULATION

70.5%

INDIVIDUALS USING
THE INTERNET



KEPIOS

**5.56
BILLION**

vs. POPULATION

67.9%

SOCIAL MEDIA
USER IDENTITIES



**5.24
BILLION**

vs. POPULATION

63.9%

SOURCES: KEPIOS ANALYSIS, UNITED NATIONS, GOVERNMENT RESOURCES, GSMA INTELLIGENCE, ITU, EUROSTAT, CNNIC, KANTAR & IAMA, PLATFORM RESOURCES, COMPANY EARNINGS REPORTS, OECD, BETA RESEARCH CENTER. **ADVISORY:** SOCIAL MEDIA USER IDENTITIES MAY NOT REPRESENT UNIQUE INDIVIDUALS. **COMPARABILITY:** BASE REVISIONS, SOURCE CHANGES. SEE [NOTES ON DATA](#).

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DIGITAL GROWTH

CHANGE IN THE USE OF CONNECTED DEVICES AND SERVICES OVER TIME



TOTAL
POPULATION



Meltwater

+0.9%

YEAR-ON-YEAR CHANGE

+70 MILLION

UNIQUE MOBILE
PHONE SUBSCRIBERS



KEPIOS

+2.0%

YEAR-ON-YEAR CHANGE

+112 MILLION

INDIVIDUALS USING
THE INTERNET



we are social

+2.5%

YEAR-ON-YEAR CHANGE

+136 MILLION

SOCIAL MEDIA
USER IDENTITIES



+4.1%

YEAR-ON-YEAR CHANGE

+206 MILLION

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SINGAPORE

OVERVIEW OF THE ADOPTION AND USE OF CONNECTED DEVICES AND SERVICES

NOTE: SIGNIFICANT REVISIONS TO SOURCE DATA MEAN THAT FIGURES SHOWN HERE ARE **NOT COMPARABLE** WITH PREVIOUS REPORTS. SEE THE IMPORTANT NOTES AT THE START OF THIS REPORT FOR DETAILS.



SINGAPORE

TOTAL
POPULATION



we
are
social

5.85
MILLION

YEAR-ON-YEAR CHANGE

+0.7%
+40 THOUSAND

URBANISATION

100%

CELLULAR MOBILE
CONNECTIONS



<O>
Meltwater

10.5
MILLION

YEAR-ON-YEAR CHANGE

+6.3%
+617 THOUSAND

TOTAL vs. POPULATION

179%

INDIVIDUALS USING
THE INTERNET



<O>
Meltwater

5.61
MILLION

YEAR-ON-YEAR CHANGE

+2.4%
+129 THOUSAND

TOTAL vs. POPULATION

95.8%

SOCIAL MEDIA
USER IDENTITIES



5.16
MILLION

YEAR-ON-YEAR CHANGE

+0.6%
+30 THOUSAND

TOTAL vs. POPULATION

88.2%

SOURCES: U.N.; GOVERNMENT AUTHORITIES; GSMA INTELLIGENCE; ITU; EUROSTAT; CNNIC; KANTAR & IAMA; PLATFORM RESOURCES; OCDH; BETA RESEARCH CENTER; KEPIOS ANALYSIS. **ADVISORY:** SOCIAL MEDIA USER IDENTITIES MAY **NOT** REPRESENT UNIQUE INDIVIDUALS. **COMPARABILITY:** SOURCE CHANGES AND BASE REVISIONS. FIGURES ARE **NOT COMPARABLE** WITH PREVIOUS REPORTS. GLOBAL DATASETS MAY USE DIFFERENT SOURCES vs. COUNTRY AND REGIONAL DATA, SO SUMS MAY NOT MATCH. **IMPORTANT:** NEGATIVE VALUES MAY INDICATE SOURCE DATA CORRECTIONS, AND **MAY NOT** REPRESENT DECREASES IN THE RELEVANT METRIC. WHERE YEAR-ON-YEAR CHANGE IS "[N/A]", COMPARISONS WITH HISTORICAL DATA WILL PRODUCE **INACCURATE RESULTS**. PLEASE SEE [NOTES ON DATA](#).

we
are
social

<O> Meltwater

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OVERVIEW OF SOCIAL MEDIA USE

HEADLINES FOR SOCIAL MEDIA ADOPTION AND USE (NOTE: USER IDENTITIES MAY NOT REPRESENT UNIQUE INDIVIDUALS)



SINGAPORE

NUMBER OF SOCIAL
MEDIA USER IDENTITIES



5.16
MILLION



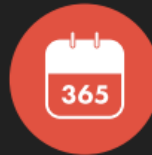
QUARTER-ON-QUARTER CHANGE
IN SOCIAL MEDIA USER IDENTITIES



0%
[UNCHANGED]



YEAR-ON-YEAR CHANGE IN
SOCIAL MEDIA USER IDENTITIES



+0.6%
+30 THOUSAND



AVERAGE DAILY TIME SPENT
USING SOCIAL MEDIA



2H 02M
YOY: -8.4% (-11 MINS)



AVERAGE NUMBER OF SOCIAL
PLATFORMS USED EACH MONTH



7.2
YOY: +4.3% (+0.3)

SOCIAL MEDIA
USER IDENTITIES vs.
TOTAL POPULATION



88.2%



SOCIAL MEDIA USER
IDENTITIES AGED 18+ vs.
POPULATION AGED 18+



91.8%



SOCIAL MEDIA USER
IDENTITIES vs. INDIVIDUALS
USING THE INTERNET



92.0%



FEMALE SOCIAL MEDIA USER
IDENTITIES vs. TOTAL SOCIAL
MEDIA USER IDENTITIES



47.6%



MALE SOCIAL MEDIA USER
IDENTITIES vs. TOTAL SOCIAL
MEDIA USER IDENTITIES



52.4%

SOURCES: KEPIOS ANALYSIS; COMPANY ADVERTISING RESOURCES; CNNIC; BETA RESEARCH CENTER; OCDH; U.N.; **GWI** (Q3 2024). **NOTE:** AVERAGE NUMBER OF PLATFORMS INCLUDES DATA FOR YOUTUBE. **ADVISORY:** SOCIAL MEDIA USER IDENTITIES MAY NOT REPRESENT UNIQUE INDIVIDUALS. COMPARISONS WITH POPULATION AND INTERNET USERS MAY EXCEED 100% DUE TO DUPLICATE AND FAKE ACCOUNTS; USER AGE MISSTATEMENTS, DIFFERENT REPORTING PERIODS, AND DIFFERENCES BETWEEN CENSUS COUNTS AND RESIDENT POPULATIONS. **COMPARABILITY:** SOURCE AND METHODOLOGY CHANGES; BASE REVISIONS. SEE [NOTES ON DATA](#).

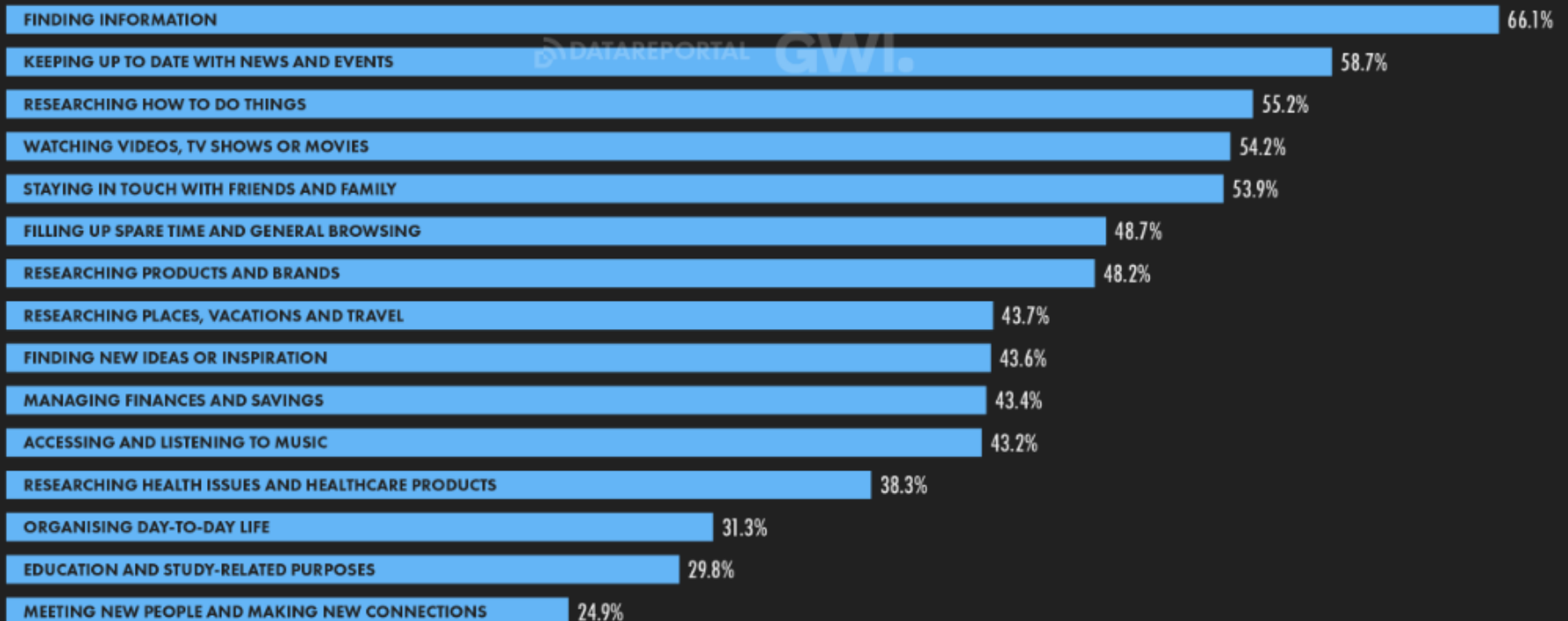
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MAIN REASONS FOR USING THE INTERNET

PRIMARY REASONS WHY **INTERNET USERS AGED 16+** USE THE INTERNET



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TOP WEBSITES: SIMILARWEB RANKING

SIMILARWEB'S RANKING OF THE MOST VISITED WEBSITES, BASED ON WEBSITE TRAFFIC BETWEEN 01 SEPTEMBER AND 30 NOVEMBER 2024



SINGAPORE

#	WEBSITE	TOTAL VISITS (MONTHLY AVE.)	UNIQUE VISITORS (MONTHLY AVE.)	AVERAGE TIME PER VISIT	AVERAGE PAGES PER VISIT
01	GOOGLE.COM	377 M	9.22 M	11M 20S	9.17
02	YOUTUBE.COM	124 M	6.02 M	18M 23S	10.69
03	INSTAGRAM.COM	28.0 M	4.77 M	8M 01S	13.43
04	FACEBOOK.COM	27.9 M	2.39 M	9M 39S	9.80
05	REDDIT.COM	27.3 M	4.91 M	6M 14S	4.16
06	CHATGPT.COM	21.3 M	1.61 M	6M 29S	3.46
07	X.COM	19.6 M	1.86 M	14M 09S	16.44
08	WHATSAPP.COM	18.7 M	3.23 M	5M 08S	2.91
09	WIKIPEDIA.ORG	15.1 M	3.26 M	3M 34S	3.34
10	SHOPEE.SG	13.4 M	3.57 M	5M 33S	6.29

#	WEBSITE	TOTAL VISITS (MONTHLY AVE.)	UNIQUE VISITORS (MONTHLY AVE.)	AVERAGE TIME PER VISIT	AVERAGE PAGES PER VISIT
11	LINKEDIN.COM	12.8 M	2.46 M	8M 25S	8.57
12	YAHOO.COM	12.1 M	1.61 M	6M 41S	4.49
13	STRAITTIMES.COM	11.9 M	1.88 M	2M 29S	2.08
14	MICROSOFTONLINE.COM	11.8 M	1.71 M	2M 33S	2.48
15	NETFLIX.COM	11.5 M	1.20 M	7M 40S	3.98
16	CHANNELNEWSASIA.COM	10.2 M	1.55 M	3M 01S	2.48
17	SINGPASS.GOV.SG	9.80 M	3.31 M	1M 53S	2.35
18	OFFICE.COM	9.15 M	852 K	9M 54S	7.42
19	TELEGRAM.ORG	8.84 M	860 K	12M 29S	7.25
20	BAIDU.COM	8.45 M	1.19 M	6M 39S	5.74

The First Thirty Seconds

- First two decades of e-commerce
 - Just the beginning
 - Rapid growth and change
- Technologies evolve at exponential rates
 - Disruptive business change
 - New opportunities
- Why study e-commerce
 - Understand opportunities and risks
 - Analyze e-commerce ideas, models, issues

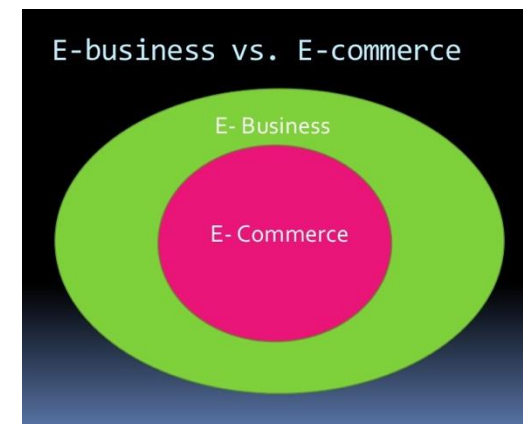
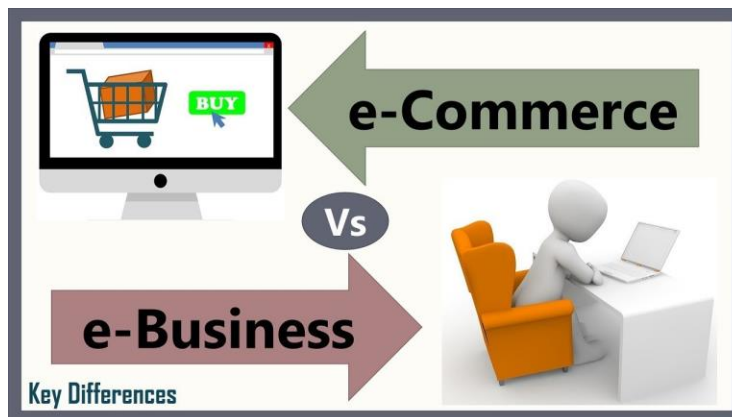
Introduction to E-commerce

- Use of Internet to transact business
 - Includes Web, mobile browsers and apps
- More formally:
 - Digitally enabled commercial transactions between and among organizations and individuals



The Difference between E-commerce and E-business

- E-business:
 - Digital enabling of transactions and processes within a firm, involving information systems under firm's control
 - Does not include commercial transactions involving an exchange of value across organizational boundaries



Technological Building Blocks Underlying E-Business

- Internet
- World Wide Web
 - HTML
 - Deep Web vs “surface” Web
- Mobile platform
 - Mobile apps

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MOBILE CONNECTIVITY

ADOPTION AND USE OF MOBILE PHONES AND DEVICES THAT CONNECT TO CELLULAR NETWORKS



GLOBAL OVERVIEW

NUMBER OF UNIQUE
MOBILE SUBSCRIBERS
(ANY TYPE OF HANDSET)



5.78
BILLION

UNIQUE MOBILE
SUBSCRIBERS AS A PERCENTAGE
OF THE TOTAL POPULATION



70.5%

ANNUAL CHANGE IN
THE NUMBER OF UNIQUE
MOBILE SUBSCRIBERS



+2.0%
+112 MILLION

CELLULAR MOBILE
CONNECTIONS
(EXCLUDING IOT)



we
are
social

8.78
BILLION

ANNUAL CHANGE IN THE
NUMBER OF CELLULAR
CONNECTIONS (EX. IOT)



+2.2%
+190 MILLION

NUMBER OF
UNIQUE CELLULAR
DATA SUBSCRIBERS



4.74
BILLION

ANNUAL CHANGE IN
THE NUMBER OF UNIQUE
CELLULAR DATA SUBSCRIBERS



+2.5%
+114 MILLION

TOTAL NUMBER OF
BROADBAND MOBILE
CONNECTIONS



8.31
BILLION

NUMBER OF BROADBAND MOBILE
CONNECTIONS AS A PERCENTAGE
OF TOTAL MOBILE CONNECTIONS



94.6%

AVERAGE NUMBER OF
MOBILE CONNECTIONS PER
UNIQUE MOBILE SUBSCRIBER



1.52

SOURCE: GSMA INTELLIGENCE. **NOTES:** CELLULAR CONNECTIONS INCLUDE DEVICES OTHER THAN MOBILE PHONES, BUT EXCLUDE CELLULAR IOT CONNECTIONS. CONNECTION FIGURES MAY EXCEED FIGURES FOR POPULATION AND UNIQUE SUBSCRIBERS DUE TO MULTIPLE CONNECTIONS PER PERSON. **COMPARABILITY:** REGULAR BASE REVISIONS. SEE **NOTES ON DATA**. **POTENTIAL MISMATCHES:** FIGURES FOR "UNIQUE CELLULAR DATA SUBSCRIBERS" MAY NOT MATCH FIGURES FOR MOBILE INTERNET USE SHOWN ELSEWHERE IN THIS REPORT, DUE TO FACTORS SUCH AS THE USE OF WIFI NETWORKS ON MOBILE DEVICES, MULTIPLE CELLULAR DATA SUBSCRIPTIONS PER PERSON, AND THE SHARING OF INDIVIDUAL CELLULAR DATA SUBSCRIPTIONS.

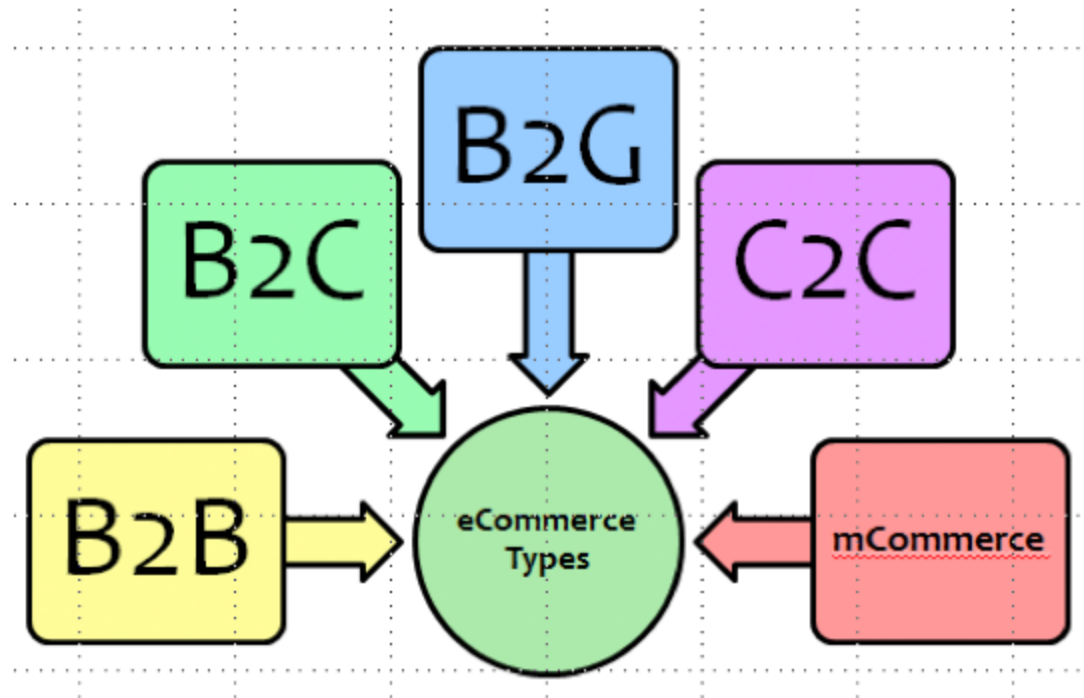
Major Trends in Digital Business

- Business trends include:
 - All forms of e-commerce show very strong growth
- Technology trends include:
 - Mobile platform has made mobile e-commerce reality
- Societal trends include:
 - Increased online social interaction and sharing

Unique Features of Digital Technology

1. Its everywhere!
2. Global reach
3. Universal standards
4. Information richness
5. Interactivity
6. Information density
7. Personalization/customization
8. Social technology

Types of E-commerce



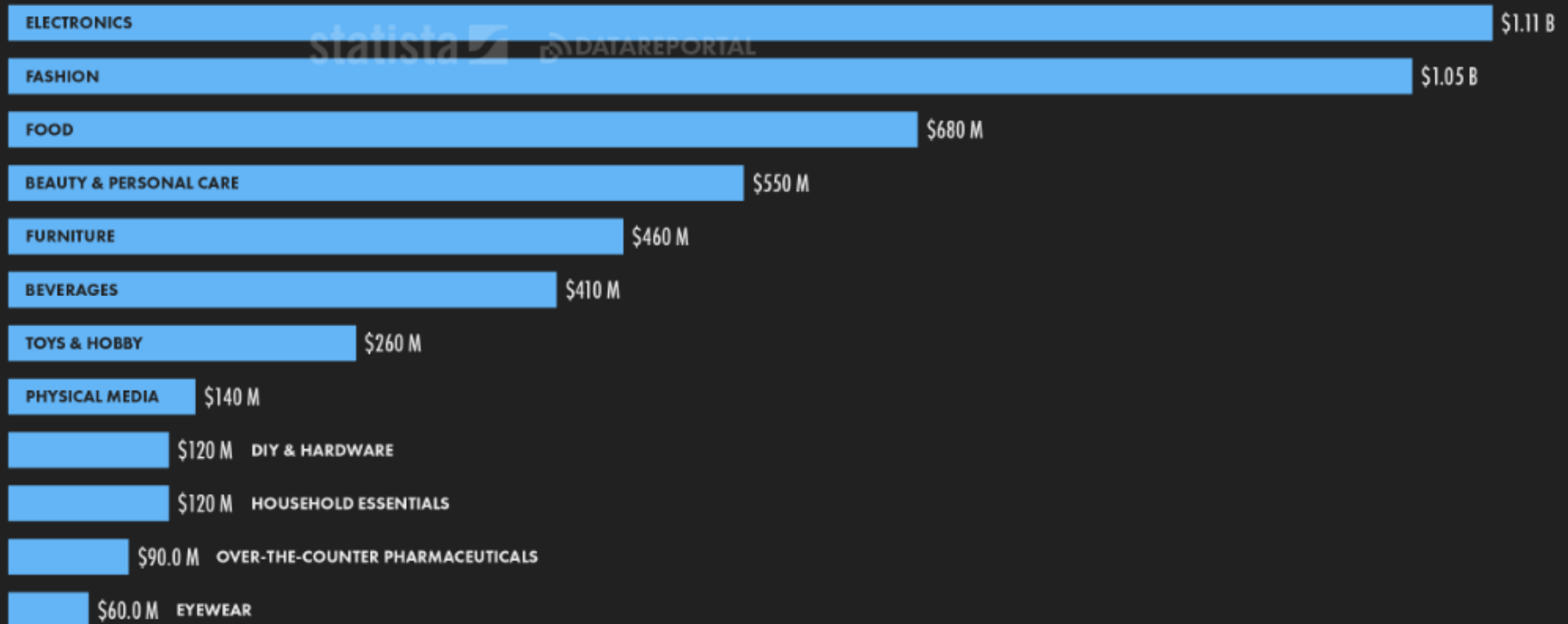
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ECOMMERCE: CONSUMER GOODS CATEGORIES

ESTIMATED ANNUAL SPEND IN EACH CONSUMER GOODS ECOMMERCE CATEGORY (B2C ONLY, U.S. DOLLARS, FULL-YEAR 2024)



SINGAPORE



Digitization: A Brief History

- Precursors
 - Baxter Healthcare modem-based system
 - Order entry systems
 - Electronic Data Interchange (E D I) standards
 - French Minitel



Digitization: A Brief History

- 1995–2000: Invention
 - Sale of simple retail goods
 - Limited bandwidth and media
 - Euphoric visions of
 - Friction-free commerce
 - First-mover advantages
 - Dot-com crash of 2000

Digitization: A Brief History

- 2001–2006: Consolidation
 - Emphasis on business-driven approach
 - Traditional large firms expand presence
 - Start-up financing shrinks
 - More complex products and services sold
 - Growth of search engine advertising
 - Business web presences expand

Digitization: A Brief History

- 2007–Present: Reinvention
 - Rapid growth of:
 - Web 2.0, including online social networks
 - Mobile platform
 - Local commerce
 - On-demand service economy
 - Artificial Intelligence
 - Entertainment content develops as source of revenues
 - Transformation of marketing (Transactional to Relationship Marketing)

Assessing The Digital World (1 of 2)

- Stunning technological success
- Early years a mixed business success
 - Few early dot-coms have survived
 - Online sales growing rapidly

Assessing The Digital World (2 of 2)

- Other surprises
 - Fast-follower advantages
 - Start-up costs
 - Impact of mobile platform
 - Emergence of on-demand e-commerce
 - The use of AI in everyday lives

Understanding Digitization: Organizing Themes

- Technology:
 - Development and mastery of digital computing and communications technology
- Business:
 - New technologies present businesses with new ways of organizing production and transacting business
- Society:
 - Intellectual property, individual privacy, public welfare policy

Questions??